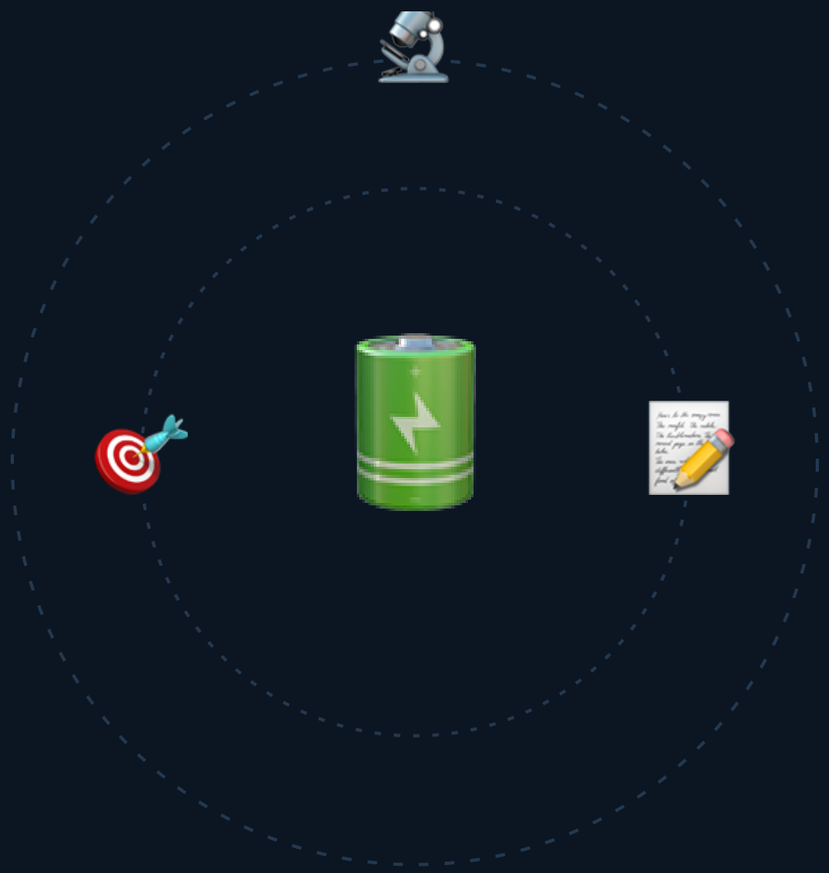
WEEK 2 · DAY 3

Lab Day + Quiz 1 + Pick Your Project



Two weeks of skills, one real investigation, one quiz, and the start of *your* project. 🚀

NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

Four moves today



Recap game

8 sharp-sharp questions

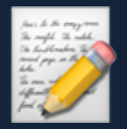
15 min



Mini investigation

is our battery getting tired?

50 min



Quiz 1

30 questions, just you

20 min



Project kickoff

pick your brief

25 min

+ 10 minutes to wrap up. **No new theory today: today you drive.** 🚗

🎮 SHARP-SHARP RECAP · ROUND 1

Shout the answer before I click!

🔀 You shuffle the rows of a daily series. Still a time series?

❌ No. The **order** carries the information. Shuffle it and the story is gone.

🕳️ An export has 1,093 rows for 1,096 days. What makes the gaps show up?

`.asfreq("D")`: one slot per day, missing days appear as **NaN**.

📉 August: 11 days had a link-down hour. How many of those days had a daily MEAN below zero?

Zero. The lowest daily mean was 4.91 dB. Report SLAs from the **minimum**, not the mean.

🗒️ Weekend NOC tickets are about half of weekday tickets. Which window smooths that out?

7 days: weekends average 3.4 tickets vs 7.1 on weekdays (48%). One window = one full week.

🎮 SHARP-SHARP RECAP · ROUND 2

Second round. Faster!

👀 Live monitoring dashboard: trailing or centred rolling window?

Trailing. A centred window peeks at readings from the future. That is “expo”. 🥷🏾

🧩 Decomposing daily NOC tickets: which period?

7, because the working week sets it. The **process** picks the period, not a default.

📊 Throughput correlogram: lag 1 = 0.58, lag 7 = 0.99. What does that tell you?

The pattern repeats **weekly**: “same weekday last week” is a very strong guess.

🧊 Amplifier temperature and RF drive correlate 0.59. Does drive cause the heat?

Not proven: after differencing it falls to **0.40**. Much of it is **shared trend**. Correlation ≠ cause.

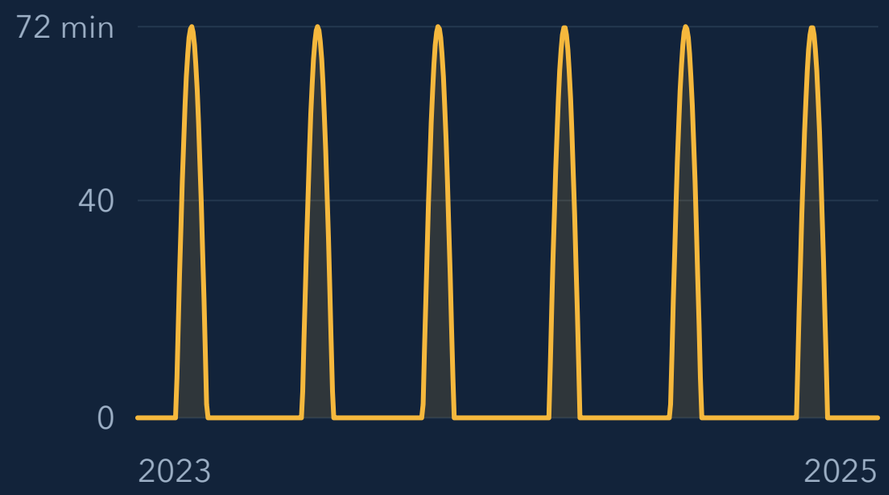
🔬 MINI INVESTIGATION · BATTERY_HEALTH_DAILY (WEEK-03 VERSION)

Your oga asks: “Is our battery getting tired?” 🔋

Estimated battery capacity (amp-hours), 3 years



Eclipse minutes per day: count the humps 👀



1,096 days

capacity 103.3 → 96.6 Ah

97 Ah = made-up line for practice

real limits come from the spacecraft engineers

THE LAB IN 7 STEPS · 50 MINUTES

Everything from Weeks 1–2, in one go



1

Load

the right file + asfreq



2

Profile

rows, gaps, duplicates



3

Plot

threshold + finding title

4

Smooth

rolling window



5

Decompose

pick the period



6

ACF

confirm the period



7

Finding

4 sentences

⚠️ The trap

This file exists in **two** original folders. Week-03 is clean; week-04 has planted faults for later. Load the right one!

🧭 Less hand-holding

Fewer hints today. Use your **rule books** and **glossaries** from Days 1–5.

HOW TO WRITE IT UP

The 4-sentence finding 📝

S **Situation:** The NOC staffs every day of the week the same way.

O **Observation:** Weekends average **3.4** tickets a day vs **7.1** on weekdays (48%).

I **Implication:** Weekend shifts carry about half the load of weekday shifts.

R **Recommendation:** Trial a lighter weekend roster with one engineer on call.

Example built from `noc_tickets_daily`. The **Observation** always carries a number.

📝 QUIZ 1 · WEEKS 1-2

30 questions. 20 minutes. Just you.

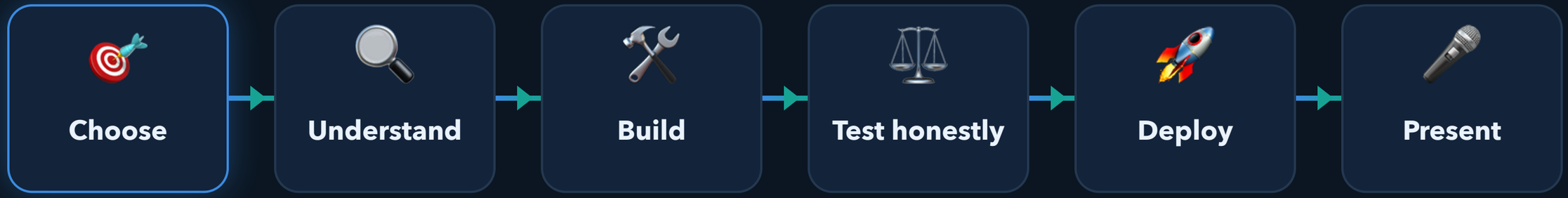
1. Phones face down, notebooks closed, **no AI button** 🙅🏾
2. Read the whole question. Don't hunt for patterns in the letters 😉
3. Not sure? Eliminate the silly options first
4. We mark it **together**: every wrong answer is a lesson

6 questions on each topic

- ⏱️ What makes a time series
- 📈 Charts that don't lie
- 〰️ Rolling windows
- 🧩 Decomposition
- 📊 Autocorrelation

🎯 FINAL PROJECT KICKOFF

One real problem. Your unit's data. A tool they can open.



👤 Individual

Your own project, your own unit.

🏢 Real data

From a named process your unit really runs.

📊 Working tool

A dashboard or daily report, on the training server.

CHOOSE ONE BRIEF

Five briefs, in plain Naija



A. Telemetry anomalies

"Which hours did the satellite behave strangely, and why?"



B. Battery & power ageing

"When will the battery be too tired to carry us through eclipse?"



C. RF link quality

"Will the rain cut our signal tomorrow?"



D. Capacity

"When will the expressway be full?"



E. Site downtime

"Which VSAT sites will give us wahala next week?"

Your process doesn't fit exactly? You can **adapt a brief** with your unit head's OK. Say what you changed.

HOW TO CHOOSE

Name a process, not a topic

❌

Too vague

“Satellite data”
“Network performance”
“Something with AI”

✅

A named process with an owner

“The weekly battery-health review on the bus subsystem”
“VSAT site fault escalation in the NOC”

1 · Is it in **my** unit?

2 · Can I get the **data**?

3 · Does someone **own** the process?

THE DATA RULES · READ THEM TWICE

Real data. Real care.



Stays inside NIGCOMSAT

Colab runs on Google's servers. Real data does **not** go there unless your unit head agrees in writing.



No raw data on GitHub

Commit your **code**, never the export. Load data from a path in a config file.



AI never sees it

Describe the **shape** of your data. Never paste restricted data into any AI tool.

Committing restricted data is a serious failure, whatever the rest of the work looks like.

THE ROAD TO DEMO DAY

Five checkpoints



WEEK 2 · DAY 3
Brief chosen
← today

WEEK 3 · DAY 3
Data in hand + profiled

WEEK 4 · DAY 3
Method drafted

WEEK 5 · DAY 3
Honest test vs baseline

WEEK 6 · DAY 3
Demo day

Today's homework: fill in the **project kickoff worksheet** and send the email to your unit head. Bring the reply to Week 3 · Day 3.

YOUR HOMEWORK, ALREADY HALF-WRITTEN

The email to your unit head ✉️

Subject: Time-series training: final-project scope for approval

Dear [Name],

For my final project I'd like to build a small monitoring tool for [named process].

- **Brief:** [A–E] • **Data:** [system, span, cadence], staying on NIGCOMSAT systems
- **Who it's for:** [team / role] • **Output:** [dashboard / daily report] on the training server

Could you confirm by replying to this email? Thank you.

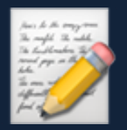
Full template: `W2D3-project-kickoff-worksheet.pdf` and the final-project requirements PDF.

TODAY IN 3 LINES

What you did today



Ran a full investigation: load → profile → plot → smooth → decompose → ACF → finding.



Quiz 1 done: Weeks 1–2 checked. Every wrong answer is tomorrow's strength.



Project started: brief picked, process named, email to your unit head on its way.

Next week 👉 **Alarms That Cry Wolf**: when is a reading actually *wrong*? 🚨